

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

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- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-a4a2a15a5c9b467ad.jsonl`  
- SHA-256: `1efaf8e9ed1f79081e600f3fae5617f70dec9da1b2042d745ab0d9b8da90b04e`  
- Source modified: `2026-06-16T03:19:28+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:17:48.908Z)

Web Searcher: Small-n evaluation rigor for video segmentation

Research question: "Research question: **What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best *recent* methods for (a) turning a dense per-frame signal into well-segmented actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMM/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k,

edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible."

Your angle: ****Small-n evaluation rigor for video segmentation**** ? Dimension 4: trustworthy measurement at n=6?16 ? bootstrap/permutation/sign-test validity, LOGO vs RL00CV bias, calibration (ECE/Brier), IAA, and multiple-comparison/over-fitting traps for a small-data leaderboard.

Search query: `small sample evaluation bootstrap permutation paired sign test leave-one-group-out cross-validation bias calibration ECE annotator agreement temporal segmentation benchmark pitfalls`

Task

Use WebSearch with the query above (or a refined version). Return the top 4-6 most relevant results.

Rank by relevance to the ORIGINAL question, not just the search query. Skip obvious SEO spam/content farms.

Include a short snippet capturing why each result is relevant.

Structured output only.

2. assistant (2026-06-16T03:17:51.084Z)

I'll research this directly using web search.

3. assistant (2026-06-16T03:17:53.874Z)

I'll search for the relevant research on small-n evaluation rigor for video segmentation.

4. user (2026-06-16T03:18:03.911Z)

Web search results for query: "small sample evaluation bootstrap permutation paired sign test leave-one-group-out cross-validation bias calibration ECE annotator agreement temporal segmentation benchmark pitfalls"

Links: [{"title": "Distributional bias compromises leave-one-out cross-validation", "url": "https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12662204/"}], [{"title": "Distributional bias compromises"}]

leave-one-out cross-validation", "url": "https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11177965/"}], [{"title": "Cross validation for model selection: A review with examples from ecology - Yates - 2023 - Ecological Monographs - Wiley Online Library", "url": "https://esajournals.onlinelibrary.wiley.com/doi/10.1002/ecm.1557"}], [{"title": "Human Psychometric Questionnaires Mischaracterize LLM Psychology: Evidence from Generation Behavior", "url": "https://arxiv.org/pdf/2509.10078"}, {"title": "Machine learning algorithm validation with a limited sample size | PLOS One", "url": "https://journals.plos.org/plosone/article?id=10.1371%2Fjournal.pone.0224365"}, {"title": "Comparing NLP Models with Confidence: The Paired Bootstrap Test Explained | by Deepankar Singh | AI-Enthusiast | Medium", "url": "https://medium.com/ai-enthusiast/comparing-nlp-models-with-confidence-the-paired-bootstrap-test-explained-c9a88532ea3d"}]]

I've completed a web search for your query. The search returned several relevant results related to evaluation methodologies and their pitfalls:

****Key findings:****

1. ****Leave-One-Out Cross-Validation Bias****: Research demonstrates that leave-one-out cross-validation creates a negative correlation between the average label of each training fold and the label of its corresponding test instance (termed "distributional bias"), which tends to negatively impact performance evaluation and hyperparameter optimization as machine learning models regress to the mean of their training data.
2. ****Leave-One-Group-Out and Temporal Structure****: When model residuals are not independent due to temporal, spatial, or other grouping structure in the data, an appropriately structured data-splitting scheme should be selected, such as blocked or leave-one-group-out cross-validation.
3. ****Small Sample Size Issues****: K-fold cross-validation produces strongly biased performance estimates with small sample sizes.
4. ****Bootstrap and Permutation Methods****: The results include discussions of bootstrap confidence intervals and sign-flip permutation tests as alternatives for statistical comparison.
5. ****Bias Correction****: For $k < 10$ in k-fold cross-validation, bias correction is usually required.

The search results address several of the key topics in your query, particularly around cross-validation pitfalls, distributional bias, and the importance of considering data structure (temporal, spatial, or grouping) when designing evaluation schemes.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

5. user (2026-06-16T03:18:20.047Z)

Web search results for query: "over-segmentation temporal action segmentation evaluation segmental F1 edit score tIoU mAP pitfalls metrics"

Links: [{"title": "Action Segmentation with Mixed Temporal Domain Adaptation", "url": "https://arxiv.org/pdf/2104.07461"}, {"title": "Video action segmentation by mixed temporal domain adaption", "url": "https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/11138441"}, {"title": "SuPRA: Surgical Phase Recognition and Anticipation for Intra-Operative Planning", "url": "https://arxiv.org/pdf/2403.06200"}, {"title": "Temporal Action Segmentation: A Comprehensive Overview", "url": "https://www.emergentmind.com/topics/temporal-action-segmentation"}, {"title": "Point-Supervised Skeleton-Based Human Action Segmentation", "url": "https://arxiv.org/pdf/2603.06201"}, {"title": "Improving Action Segmentation via Graph Based Temporal Reasoning", "url": "https://openaccess.thecvf.com/content_CVPR_2020/papers/Huang_Improving_Action_Segmentation_via_Graph-Based_Temporal_Reasoning_CVPR_2020_paper.pdf"}, {"title": "Language-Assisted Human Part Motion Learning for Skeleton-Based Temporal Action Segmentation", "url": "https://arxiv.org/pdf/2410.06353"}, {"title": "An Efficient Framework for Few-shot Skeleton-based Temporal Action Segmentation", "url": "https://arxiv.org/pdf/2207.09925"}, {"title": "Long-Tail Temporal Action"}]

Segmentation with Group-wise Temporal Logit
Adjustment", "url": "https://arxiv.org/html/2408.09919v1"}]

Based on the search results, here's a comprehensive overview of temporal action segmentation evaluation metrics and their pitfalls:

Key Evaluation Metrics

Common evaluation metrics for action segmentation include frame-wise accuracy (Acc), segmental edit score, and segmental F1 score at IoU thresholds (F1@k, where $k = \{10, 25, 50\}$).

Limitations and Pitfalls

Frame-wise Accuracy Limitations

Frame-wise accuracy does not account for temporal dependencies and causes large qualitative differences, while long action classes have higher impact than shorter ones, making this metric unable to reflect over-segmentation errors.

Segmental Edit Score

The segmental edit score penalizes over-segmentation by measuring the ordering of predicted action segments independent of slight temporal shifts. Edit Score evaluates alignment of predicted segment boundaries with ground truth, is particularly adept at handling over-segmentation errors where a continuous action is erroneously divided into multiple segments, and is less sensitive to such fragmentation.

Segmental F1 Score (F1@IoU/tIoU)

Segmental F1 score (F1@k) has become popular since it better indicates qualitative segmentation results and penalizes over-segmentation errors while ignoring minor temporal shifts between predictions and ground truth. F1@k is determined by the total number of actions but not the duration of each action instance, which is similar to mean average precision (mAP) with intersection-over-union (IoU) overlap criteria.

Why Segment-Based Metrics Matter

Temporal segmentation is prone to over-segmentation where an action is fragmented into multiple discontinuous segments, making frame-wise accuracy alone inadequate, whereas Edit and F1 scores are more suitable for evaluating such phenomena.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-06-16T03:18:23.018Z)

Let me dig deeper into the most relevant specific topics: small-data calibration/ECE bias, IAA for temporal segmentation, and the over-segmentation metric foundations.

7. user (2026-06-16T03:18:31.589Z)

Web search results for query: "calibration error estimation bias finite samples ECE estimator variance small dataset reliable Brier score evaluation"

Links: [{"title": "Estimating Expected Calibration Error for Positive-...", "url": "https://openreview.net/attachment?id=SvoBtLIrPZ&name=pdf"}, {"title": "Mitigating bias in calibration error estimation", "url": "https://www.arxiv.org/pdf/2012.08668v1"}, {"title": "Mitigating Bias in Calibration Error Estimation", "url": "https://www.researchgate.net/publication/347398268_Mitigating_bias_in_calibration_error_estimation"}, {"title": "Expected Calibration Error (ECE) Overview", "url": "https://www.emergentmind.com/topics/expected-calibration-error-ece"}, {"title": "Mitigating Bias in Calibration Error Estimation", "url": "https://arxiv.org/pdf/2012.08668"}, {"title": "Information-theoretic Generalization Analysis for Expected Calibration Error", "url": "https://arxiv.org/pdf/2405.15709"}, {"title": "MITIGATING BIAS IN CALIBRATION ERROR ESTIMATION", "url": "https://openreview.net/pdf?id=NgZKCRKaY3J"}, {"title": "Revisiting the"}]

Calibration of Modern Neural Networks", "url": "https://arxiv.org/pdf/2106.07998"}]

I've found relevant research on calibration error estimation bias in finite samples. Here are the key findings:

ECE Estimator Bias and Finite Samples

The binned ECE (ECE_bin) has non-negligible statistical bias and systematically predicts large errors for perfectly calibrated models. ECE_bin can systematically underestimate or overestimate the true calibration error depending on the nature of model miscalibration, the size of the evaluation data set, and the number of bins.

Bias is highest when the model is perfectly calibrated and generally decreases as true calibration error increases, and while a larger sample size of $n=5,000$ reduces the bias, with perfect calibration ECE_BIN can still be off by 2%.

Sources of Bias

There are two sources of bias: from estimating ECE by binning, and from the finite sample size used to estimate the per-bin statistics. There is a significant gap in comprehensive theoretical understanding of binning bias and statistical bias resulting from finite test data samples.

Variance and Reliability Issues

Reliable, low-variance estimation on finite samples may not be possible for all candidate forecasters, and statistical properties like bias or consistency depend on unknown aspects of the prediction distribution.

Alternative Approaches

The monotonic sweep calibration error (ECE_sweep) produces a less biased estimator of calibration error, and kernel smoothing with a reflected Gaussian kernel yields a continuous, stable calibration error estimate. Alternative calibration metrics include negative log-likelihood and Brier score.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. user (2026-06-16T03:18:32.906Z)

Web search results for query: "inter-annotator agreement temporal boundaries event segmentation video Cohen kappa Krippendorff alpha boundary tolerance reliability"

Links: [{"title": "+ CS140 February 28, 2017 Inter annotator agreement", "url": "https://www.cs.brandeis.edu/~cs140b/CS140b_slides/CS140_Lect_7_InterAnnotatorAgreement.pdf"}, {"title": "Dense-Captioning Events in Videos", "url": "https://arxiv.org/pdf/1705.00754"}, {"title": "Counting on Consensus: Selecting the Right Inter-annotator Agreement Metric for NLP Annotation and Evaluation", "url": "https://arxiv.org/html/2603.06865"}, {"title": "Krippendorff's Alpha for Annotation Agreement | Label Studio", "url": "https://labelstud.io/blog/how-to-use-krippendorff-s-alpha-to-measure-annotation-agreement/"}, {"title": "MM-CoT: A Benchmark for Probing Visual Chain-of-Thought Reasoning in Multimodal Models", "url": "https://arxiv.org/pdf/2512.08228"}, {"title": "On User Interfaces for Large-Scale Document-Level Human Evaluation of Machine Translation Outputs", "url": "https://arxiv.org/pdf/2104.10408"}, {"title": "Calculate inter-annotator reliability", "url": "https://www.mpi.nl/tools/elan/docs/manual/Sec_Calculate_inter-annotator_reliability.html"}, {"title": "Decomposing and Recomposing Event Structure", "url": "https://arxiv.org/pdf/2103.10387"}, {"title": "Segmentation Similarity and Agreement", "url": "https://arxiv.org/pdf/1204.2847"}, {"title": "HomeSafe-Bench: Evaluating Vision-Language Models on Unsafe Action Detection for Embodied Agents in Household Scenarios", "url": "https://arxiv.org/pdf/2603.11975"}]

I've completed the web search for your query about inter-annotator agreement in temporal

boundary event segmentation. Here are the key findings:

Relevant Reliability Measures

Common inter-annotator agreement metrics include Cohen's Kappa, Fleiss's Kappa, Scott's κ , and Krippendorff's α .

Key Metrics for Your Topic

Cohen's Kappa: Cohen's kappa measures annotator agreement while accounting for chance agreement, with values close to 0 indicating no agreement and $\kappa = 1$ indicating perfect agreement. For temporal annotations, two annotators can be considered in agreement if their assigned scores differ no more than a predefined tolerance.

Krippendorff's Alpha: Compared to other agreement metrics like Cohen's kappa, Krippendorff's alpha offers more flexibility and is better suited for real-world data annotation workflows.

Boundary Tolerance in Video Event Segmentation

In video event captioning studies, temporal agreement between annotators can be measured using temporal Intersection over Union (tIoU), with workers generally agreeing when annotating temporal boundaries of video events.

Segmentation-Based Metrics: Structured and segmentation-based metrics such as F1, Boundary Similarity, Edit Distance, WindowDiff, and Pk lack standard interpretive scales, with scores depending heavily on task setup, annotation granularity, and tolerance for partial matches.

Handling Temporal Boundaries

For temporal keyframes, researchers use metrics like Lin's CCC, ICC, and MAE, and samples exhibiting temporal disparities beyond predefined tolerance thresholds are often re-annotated to establish consensus ground truth.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-06-16T03:18:35.669Z)

I have strong candidates. Let me run two more targeted searches to confirm the over-segmentation-blindness of plative tIoU and small-n significance-test validity/multiple-comparison issues.

10. user (2026-06-16T03:18:46.022Z)

Web search results for query: "segmentation similarity boundary edit distance metric near-miss boundaries temporal segmentation evaluation Fournier Inkpen"

Links: [{"title": "[PDF] Evaluating Text Segmentation using Boundary Edit Distance | Semantic Scholar", "url": "https://www.semanticscholar.org/paper/Evaluating-Text-Segmentation-using-Boundary-Edit-Fournier/d613425ab2c9a0b8b7905ba4ce732a8da8733edf"}, {"title": "140 acl-2013-Evaluating Text Segmentation using Boundary Edit Distance", "url": "https://makerhacker.github.io/paper-mining/acl/acl2013/acl-2013-Evaluating_Text_Segmentation_using_Boundary_Edit_Distance.html"}, {"title": "(PDF) Evaluating Text Segmentation using Boundary Edit Distance", "url": "https://www.researchgate.net/publication/237073700_Evaluating_Text_Segmentation_using_Boundary_Edit_Distance"}, {"title": "Segmentation Similarity and Agreement | PPT", "url": "https://www.slideshare.net/cfournie/segmentation-similarity-and-agreement"}, {"title": "[1204.2847] Segmentation Similarity and Agreement", "url": "https://arxiv.org/abs/1204.2847"}, {"title": "Segmentation Similarity and Agreement", "url": "https://arxiv.org/pdf/1204.2847"}, {"title": "segmentation.evaluation/docs/index.rst at main · cfournie/segmentation.evaluation", "url": "https://github.com/cfournie/segmentation.evaluation/blob/main/docs/index.rst"}, {"title": "ACL 2013 - Evaluating Text Segmentation using Boundary Edit Distance", "url": "https://mirror.aclweb.org/acl2013/accepted/458.html"}]

Based on the search results, here's what I found regarding the terms in your query:

Key Findings

Fournier and Inkpen proposed a new segmentation evaluation metric called segmentation similarity (S), which quantifies the similarity between two segmentations as the proportion of boundaries that are not transformed when comparing them using edit distance, using edit distance as a penalty function and scaling penalties by segmentation size.

Existing segmentation metrics such as Pk, WindowDiff, and Segmentation Similarity (S) are able to award partial credit for near misses between boundaries, but are biased towards segmentations containing few or tightly clustered boundaries.

Related Work

A subsequent work proposes a new segmentation evaluation metric named boundary similarity (B), along with an inter-coder agreement coefficient adaptation and a confusion-matrix for segmentation, all based upon an adaptation of the boundary edit distance in Fournier and Inkpen (2012).

Segmentation Similarity (S) has three parameters: n (the number of potential boundaries considered a near miss, with a default of 2), TE (to use transposition error scaling), and weights upon error types to reduce their severity.

The research demonstrates how edit distance metrics can be adapted to evaluate text segmentation while handling near-miss boundaries appropriately.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

11. user (2026-06-16T03:18:48.279Z)

Web search results for query: "permutation test versus bootstrap small sample size validity exact paired test statistical power machine learning model comparison multiple comparisons correction"

Links: [{"title": "(PDF) Bootstrapping and permuting paired t-test type statistics", "url": "https://www.researchgate.net/publication/257665537_Bootstrapping_and_permuting_paired_t-test_type_statistics"}, {"title": "A Comparison of Permutation and Bootstrap Tests", "url": "https://math.ucsd.edu/sites/math.ucsd.edu/files/undergrad/honors-program/honors-theses/2016-2017/He_Jiang_Honors_Thesis.pdf"}, {"title": "A Statistical Analysis of Summarization Evaluation Metrics using Resampling Methods", "url": "https://arxiv.org/pdf/2104.00054"}, {"title": "A Hitchhiker's Guide to Statistical Comparisons of Reinforcement Learning Algorithms", "url": "https://arxiv.org/pdf/1904.06979"}, {"title": "Chapter 4 Resampling: permutation testing and bootstrap | Computational Statistics I", "url": "https://www.cs.helsinki.fi/u/ahonkela/teaching/compstats1/book/resampling-permutation-testing-and-bootstrap.html"}, {"title": "Analysis of small sample size studies using nonparametric bootstrap test with pooled resampling method - Dwivedi - 2017 - Statistics in Medicine - Wiley Online Library", "url": "https://onlinelibrary.wiley.com/doi/10.1002/sim.7263"}, {"title": "Extremely efficient permutation and bootstrap hypothesis tests using R", "url": "https://arxiv.org/pdf/1806.10947"}]

Based on the search results, here's what research shows about permutation tests and bootstrap methods in the context you're interested in:

Small Sample Size Validity

Permutation tests are exact tests, meaning that if a test is set at α level, the Type I Error of the permutation test will be less than or equal to α . However, the bootstrap test should never be used for sample sizes below $N=50$ and the permutation test should never be used for sample sizes below $N=10$. The bootstrap test uses the sample as an estimate of the true performance

distribution, and a small sample is a very noisy estimate, which leads to very high false positive rates.

Paired Testing

For matched pairs where data may not be exchangeable under the null, permutation approaches are investigated as valid procedures for finite sample sizes, and these schemes are asymptotically valid. Permutation tests tend to be used more frequently than paired bootstrap tests for hypothesis testing because they directly test whether any observed difference between two values is due to random chance.

Statistical Power

Permutation and bootstrap schemes can improve the power of the t-test under non-normality. For machine learning model comparison, nonparametric pooled bootstrap t-tests provided equal or greater power for comparing two means compared with unpaired t-tests, Welch t-tests, Wilcoxon rank sum tests, and permutation tests while maintaining type I error probability.

The sources don't extensively address multiple comparisons corrections in this context, but the validity considerations differ significantly between methods depending on sample size.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

12. assistant (2026-06-16T03:18:51.058Z)

I have enough high-quality, verified sources covering all of Dimension 4. Let me compile the top results, ranked by relevance to the original question (small-n evaluation rigor for a badminton rally-segmentation leaderboard).

13. user (2026-06-16T03:19:17.936Z)

Structured output provided successfully

14. assistant (2026-06-16T03:19:28.588Z)

I've completed the research on Dimension 4 (small-n evaluation rigor) and returned six ranked, verified sources via StructuredOutput. Summary of what they cover and how they map to the project's existing harness:

- ECE bias at small n (Roelofs et al., AISTATS 2022) ? the binned ECE the harness already computes is biased even at n=5000; at 6-16 videos it should not gate decisions without debiasing or falling back to Brier/NLL.
- Bootstrap/permutation validity (Dwivedi 2017 + resampling-floor finding) ? bootstrap is unreliable below ~N=50, permutation below ~N=10; at n=6 the paired exact sign-test is the trustworthy primary, confirming the harness's conservative design.
- LOGO vs RL00CV bias (distributional-bias L00CV paper) ? validates grouping leave-one-video-out by match and warns the tiny-n LR scorer's tuning is itself biased toward the train mean.
- Boundary/Segmentation Similarity (Fournier & Inkpen) ? boundary-edit-distance metrics that give partial credit for near-misses AND a matching inter-coder agreement coefficient, addressing both tIoU's over-merge blindness and golden-set IAA in one framework.
- TAS metric overview ? crystallizes why segmental F1@k + edit/Levenshtein + segment-count-ratio (already in the harness) catch the mega-window over-merge bug that frame accuracy/raw tIoU miss.
- IAA metric selection (Counting on Consensus, 2026) ? guidance for choosing chance-corrected agreement vs tolerance-banded segmentation agreement for the Roora golden labels.

All URLs were returned by live search; I did not fetch full PDFs, so exact reported numbers beyond those quoted in the search abstracts (e.g., the "~2% ECE at n=5000" and "N=50/N=10 resampling floors") are as-stated in the result snippets and should be confirmed against the source PDFs before being cited as hard figures.